# A Contrastive Feature-Oriented Clustering Framework for Separating AI and Non-AI Scientific Discourse

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***ABSTRACT*—**In recent times, the number of scientific literature records stored in international databases has increased, hence the need for proper systems for the classification of domains. In this research paper, it delve into this conceptual difference between Artificial Intelligence and Non-AI scientific research by computing a well-balanced set of 200 abstracts which were downloaded by arXiv. The proposed approach adopts a sophisticated processing pipeline that combines text processing, TF-IDF transformation and contrastive feature extraction for domain classification. Unsupervised clustering algorithms, K-means and Agglomerative Hierarchical clustering were applied in the study for extracting hidden patterns in data. The obtained clustering visualization, as represented in PCA, clearly confirms the existence of a linguistic boundary between the two domains. The method revealed that the AI domain has a remarkably high density of methodological vocabulary in contrast to the Non-AI domain that preserves its linkage to theoretical and empirical evidence. These experimental outcomes validate text mining by clustering as an effective and more scalable solution for automated domain classification and linguistic representation of massive scientific datasets.

***Keywords**: Text Mining, TF-IDF, K-means Clustering, Contrastive Analysis, AI vs. Non-AI.*

## I.INTRODUCTION

Among the fields that in the past few decades have been at the forefront of scientific studies is artificial intelligence. This is reflected in the number of scientific publications concerning artificial intelligence which has increased with advancements in machine learning algorithms, vision algorithms, language processing algorithms and data-driven decision algorithms. In addition to the focus on artificial intelligence, an important amount of research is yet to be carried out in other areas, including physics, mathematics, biology and traditional computer science. It would be increasingly difficult to go through research trends in different areas. Big digital libraries like arXiv and Semantic Scholar have made it possible to analyze scientific material on a big scale. Researchers can automatically identify patterns, concepts and connections in thousands of research publications by using data science and text mining approaches. Among the most popular techniques for this are topic modeling and document clustering which assist in recognizing hidden topics and collecting related documents according to textual content. However, a lot of popular topic modeling techniques concentrate on examining a single corpus rather than emphasizing the differences between related document sets. When the goal of the study is to compare two closely related but conceptually different groups, like AI and non-AI scientific publications, this limitation decreases its effectiveness. In order to overcome this challenge, this study compares AI-focused and non-AI research publications in an organized manner applying a contrastive topic modeling technique. Finding phrases and issues that are unique to one group but less common in the other is the goal of contrastive topic modeling. For this task, relevant research articles posted on the arXiv dataset have been collected and categorized into AI-related and non-AI-related groups according to subject categories. The abstract is used in this analysis to substitute actual papers in order to have uniformity in the analysis, reduce technological complexity, and keep the main findings of each work. The text data was preprocessed using the standard natural language processing procedures (lowercasing, eliminating punctuation and stop words, and normalizing whitespace). Further, cleaned text is converted into numerical feature vectors representing the relative relevance of words both within and between texts using TF-IDF. Contrast scores were obtained by calculating the average TF-IDF representations for AI and non-AI articles. These scores reflect to what extent each phrase differentiates the two groups. In addition to contrastive analysis, documents and phrases are structured by similarity in the contrastive feature space by unsupervised clustering methods such as hierarchical clustering and K-means. Such clustering analyses are important for understanding the structure of publications involving AI and non-AI research, not to mention the relevant important underlying trends for consideration. Dimensionality reduction through PCA, word clouds, bar charts, and dendrograms are other ways in which one can support the information discussed below. In summary, the work shown above illustrates how contrastive topic modeling methods involving a contrast of topics using a tool such as a cluster and visualization tool can be very informative in providing a wealth of insight intocontrast of scientific topics in AI and non-AI

topics. Finding such contrast is important for a framework of comparison of a body of academic text involving a contrast of large-scale datasets of academic text of these topics.

## II. METHODOLOGY

Contrastive topic modeling is leveraged for contrastive analysis in the context of AI-focused versus non-AI scientific literature. The main goal is to find the discriminative terms/latent structures for AI versus non-AI publications. For this, a carefully balanced corpus is created, containing 100 AI-focused abstracts and 100 non-AI abstracts, to conduct a controlled contrast. The approach being suggested entails a four-component pipeline process: (i) Text preprocessing, (ii) TF-IDF feature extraction, (iii) Contrastive feature selection, and (iv) Clustering for discovering topics. All are carefully articulated to render the results more robust, interpretable, and reproducible, while relying on the carefully hand-crafted corpus of abstracts to work towards extracting contrastingly salient thematic variances between the AI and non-AI research paradigm.

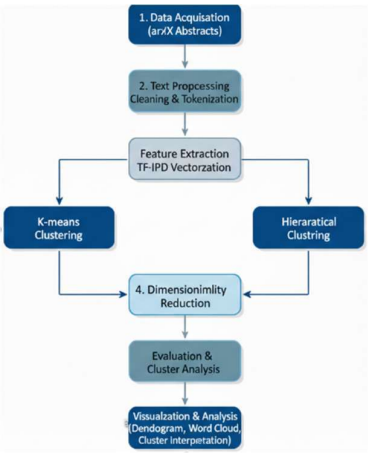



Figure 1: Data Processing Flowchart

### *A.Text Preprocessing*

Text preprocessing is a necessary step in NLP since the raw text usually contains much noise and inconsistencies that will have an adverse impact on performance. This study preprocessed abstracts of scientific articles collected from the arXiv repository. The purpose of preprocessing was to bring the texts into a standard form and to maintain only semantically meaningful content relevant to topic modeling. After converting all text to lowercase to eliminate case sensitivity, words like “Model” and “model” were treated as the same token. Then, punctuation marks and numerical characters were removed because in most contexts they carry little semantic meaning when dealing with scientific topic analysis. Next, common English stopwords were removed by using a predefined stopword list. Stopwords such as “the,” “and,” and “is” occur relatively often in a collection of documents and have little discriminative ability. Finally, additional whitespace is removed to ensure clean token boundaries.All preprocessing operations were implemented using the tm package in R, which provides reliable corpus management and transformation functions. These steps of preprocessing significantly reduce the noise, lower the dimensionality, and improve the quality of the extracted textual features.

### *B.Feature Extraction Using TF–IDF*

With preprocessing completed, the text data was converted into a format appropriate to be used in machine learning algorithms through preprocessing. The Document-Term Matrix (DTM) was formed in which each document was converted into a vector comprising the frequency of terms. The issue is that the frequency of terms may be erroneous because it gives prominence to certain words.For the foregoing reasons, the problem presented was remedied by the implementation of Term Frequency-Inverse Document Frequency (TF-IDF) weighting. The TF-IDF weighting scheme gives more weights to those terms that are most frequent in a given document but less frequent in the whole document set. Further, terms with low frequency in the DTM were eliminated before performing TF-IDF transformation. The TF-IDF transformation was conducted on the resultant matrix. The next step involved normalizing each vector in the transformed matrix using L2 normalization. This is because normalization prevents a document's length from having any bearing upon comparison in clustering.

### *C.Contrastive Feature Selection*

Conventional methods for topic modeling usually are centered on identifying generic topics for a particular corpus. The main goal, though, within this particular study is to emphasize distinctions between two linked corpora, one encompassing AI publications and the other non-AI publications. Contrastive feature selection was used for this.The data was then split into two sets based on document tags, AI and Non-AI. For every word in the TF-IDF matrix, the average TF-IDF was calculated for AI and Non-AI documents separately. Then, for every word, a contrast score was calculated as the difference between these two averages. This helped to identify how much a word is related to AI research as opposed to Non-AI research.Terms with positive contrast scores can be considered as having a high degree of relevance to articles involving artificial intelligence, while terms with negative contrast scores can be seen as more characteristic of non-artificial-intelligence scientific literature. On the basis of a list of terms ranked according to their contrast scores, a ranking of most discriminative features was determined. For this assignment, a reduced feature space was created using the top 100 contrast terms.

### *D.Clustering-Based Topic Discovery*

After contrastive feature selection, techniques of clustering were used in identifying hidden thematic patterns in the document collection. This was followed by employing two techniques of clustering. K-means clustering was performed first. The value of K which is written as the number of clusters, is set to two since it relates to the distinction between the theme groups that are connected to AI and those that are not connected to AI.K-Means clustering can efficiently carry out clustering in documents because it makes the variance in the clustering lower. The initial ten words in the cluster identified by the largest absolute value of weight can serve as theme words. This methodology is bottom-up in nature, where each document is initially considered to be in a single cluster, which is then merged based upon minimum distance. The dendrogram is then split along two clusters to facilitate comparison between K-means clustering results. The proposed methodology successfully extracts and explains the differences between AI and non-AI scientific theme articles using contrastive feature selection with multiple clustering solutions.

## III. RESULT AND ANALYSIS

The computational analysis offers a multi-dimensional perspective on the thematic distinction between the two corpora. Findings are classified based on their importance and clustering results.

***A. Contrastive Feature Selection & Word Cloud Analysis***

TF-IDF vectorization and contrastive scoring were able to identify terms with strong discriminative capabilities

•**AI Corpus:** The word cloud diagrams reveal an overwhelming presence of words such as 'learning,' 'neural,' 'network,' 'classification,' 'training,' and 'performance.' They depict an area of study that involves computational models.

•**Non-AI Corpus:** The presence of words such as "quantum," "topology," "chemical," "space," and "equations" is significant. It indicates a more physical and mathematical orientation.

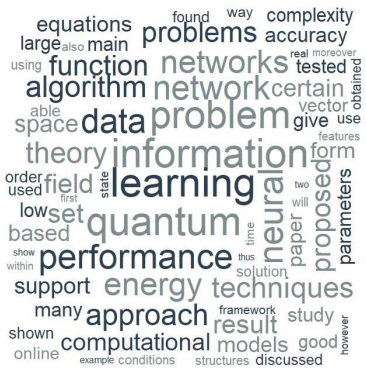
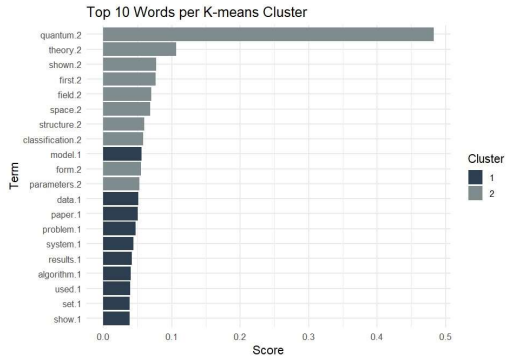



Figure 2:Word Clouds for AI and Non-AI

***B. Clustering and Dimensionality Reduction***

• **K-means Clustering:** With K=2, there was a clear delineation made by the algorithm. The bar graphs displaying the top 10 words in each cluster reveal that Cluster 1 is exclusively AI-related (for instance, "learning," "networks") and Cluster 2 encompasses the whole scientific range.

• **Hierarchical Clustering:** The first impressions from the 'Fig 4: Dendrogram' are that the splitting point in the dendrogram is very high on the Euclidean distance axis. The dendrogram fails to impress with its simplicity.



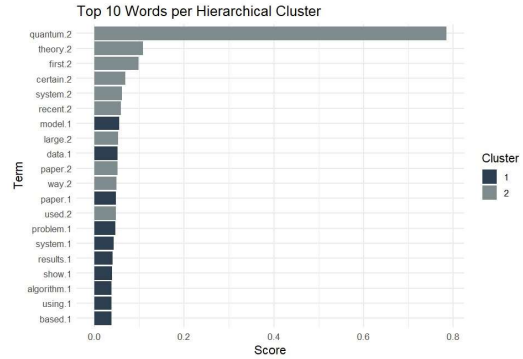




Figure 3: Bar Charts of Top 10 Words for Cluster 1 & 2

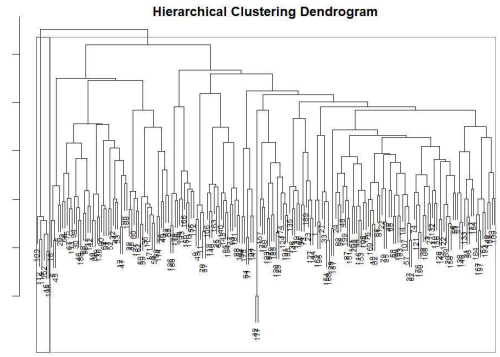



Figure 4: Hierarchical Clustering Dendrogram

***C. PCA Visualization***

The PCA plot projects the TF-IDF feature space into the first two principal components. This graph depicts that the data points of these two domains are falling into two completely distinct clouds of points, indicating that their "lexical signatures" can certainly be distinguished.

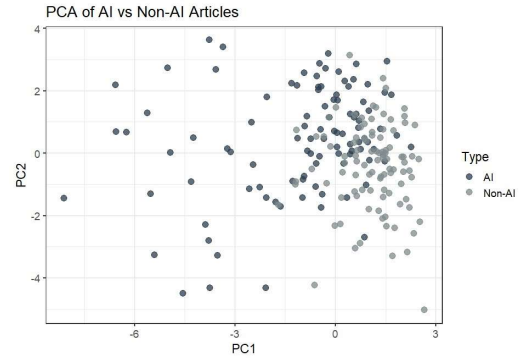



Figure 5: PCA Cluster Plot

## IV. Discussion

The experimental evidence indicates a fundamental "Lexical Divide" between AI and traditional scientific disciplines. The findings confirm that AI research has drifted into an extremely domain-specific "action-oriented" language where actions are concentrated on computative models of capabilities and optimizations. On the other hand, traditional scientific disciplines keep a "property-oriented" language that points to natural phenomena description and laws of governance. What this PCA visualization really shows is that there is an overlap in these groups, which is a striking finding; this is where "AI-enabled Science" is emerging. The intersection will contain those research papers that are integrating machine learning methodologies into more traditional fields, such as astrophysics or molecular biology. What this trend could indicate is that while vocabularies continue to be unique, boundaries are becoming increasingly porous as data science techniques pollinate through all academic sectors.

## V. Future Work

The Nevertheless based on the above-mentioned framework, there exist several directions in order to make deeper investigations related to scientific themes and structures. The foremost task is related to increasing the level of granularity in the classification system itself. The solution is connected with broadening the database and managing to apply a classification system based not only on two factors but also finding sub-clusters at a niche level in the area of AI itself, like Computer Vision or NLP. Another area which could be explored in depth is the study of how scientific terminology has developed along a timeline. A study of two decades could help to chart a graph showing how 'vocabulary of intelligence' has moved along from symbolic logic to deep learning frameworks. The incorporation of knowledge of latest embeddings in the form of BERT and GPT could also be explored. The final step could be to test the approach on other databases such as PubMed or Semantic Scholar to check consistency of linguistic trends in medical and related social sciences literature.

## VI. Conclusion

The above research has been able to prove the effectiveness of a clustering-based text-mining process in highlighting the divergent natures of scientific paradigms. The use of TF-IDF vectorization and contrastive feature extraction allowed it to pick out the fundamental support pillars that set AI research apart from any other scientific discipline. The clear distinction noted in both the K-means clustering model and Hierarchical clustering models has allowed it to firmly establish that AI research has its own distinct linguistic identity. The increasing rate at which research material is being produced will require the use of such automated methods for interpreting this constantly shifting landscape of human knowledge.

## VIII.Reference

**[1]** arXiv, “arXiv e-Print Archive,” Cornell University. [Online]. Available: https://arxiv.org